

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)		The highest levels (of NO ₂) are {along major roads / where there are most vehicles / owtte} Accept specifically named roads			1	1		
		(ii)	I	circle on black rectangle dot on M4 near Port Talbot			1	1		
			II	the level (of NO ₂ most) exceeds WHO limits / values >60 compared with {WHO limit / 40}			1	1		
	(b)	(i)		• Critical value is 1.701 (at p=0.05) (1) • Because t value is greater than {critical value/ 1.701} / 1.769> 1.701 / ORA (1) • Reject null hypothesis (1) • There is a <u>significant</u> difference between the <u>means</u> / the difference between the <u>means</u> is not due to chance (1) Ecf from incorrect critical value		3	1	4	4	
		(ii)		Emissions were (significantly) higher after closure / ORA (1) Keeping junction open will result in {lower NO ₂ levels / better air quality} (1) Ecf from (i) There was no significant difference in (mean) emissions (1) Closing the junction will not result in any changes in the NO ₂ levels / air quality} (1)			2	2		

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		(iii)	I	- Trees absorb more NO₂ {between {March / April} and {July / August} / when the road was open} / Trees absorb less NO₂ {after August / when the road was closed} (1) - This reduces confidence in the conclusion that closing the road increased NO₂ (1)			2	2		
			II	(Repeat the closure experiment in) {the other months / over longer time period / different times of the year}			1	1		
	(c)			Nitrobacter (1) (Converts nitrite) to {nitrate / NO ₃ ⁻ } (1) (Nitrate) can be {absorbed / converted into amino acids} by {plants / producers} / broken down by {denitrification / <i>Pseudomonas</i> } (1)	2	1		3		
				Question 3 total	2	4	9	15	4	0